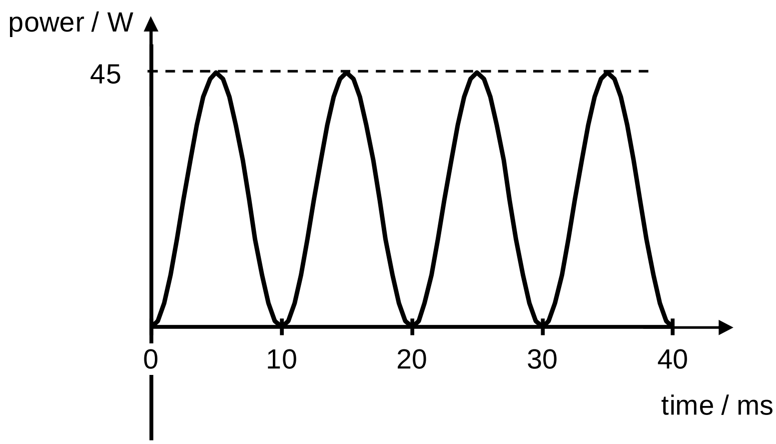
	(do not accept circular path)	description
6(a)	The root-mean-square value of an alternating current is the value of steady direct current which would produce the same mean power as the alternating current in a given resistance.	[1] answer
6(b)(i)	Period, T = 20 ms Frequency, $f=\frac{1}{T}=\frac{1}{20\times10^{-3}}=50\text{ Hz}$	[1] answer
6(b)(ii)	Peak value = 3.0 A	[1] answer
6(b)(iii)	Root-mean-square value $=\frac{\text{peak value}}{\sqrt{2}}$ $=\frac{3.0}{\sqrt{2}}$ $=2.1\text{ A}$	[1] answer
6(c)	 Peak power, $P_0=I_0^2R$ $=3.0^2\times5.0$ $=45\text{ W}$	[1] correct graph [1] correct value of peak power

No.	Solution	Remarks
6(d)	$\frac{I_s}{I_p} = \frac{N_p}{N_s}$ $\frac{I_s}{3} = \frac{300}{6000}$ $I_s = 0.15$ A	 [1] substitution [1] answer
7(a)(i)	A photon is a quantum of electromagnetic radiation with energy	[1] for correct